Two-Machine Power System with a Grid-Forming Converter

Governing Equations and Their Derivation, in the Order of the Simulink Models (Cases 1–4)

The test system, operating point and parameters are those of [1, 2]; the modelling follows [3, 4].

Study cases. The equations below are those of the complete model `case3_primary_AGC_GFM.slx`. The other cases are separate model files from which blocks are removed:

Case	Model file	AGC	GFM	States
1	`case1_primary_control.slx`	–	–	18
2	`case2_primary_AGC.slx`	✓	–	20
3	`case3_primary_AGC_GFM.slx`	✓	✓	23
4	Case 3 model run with $K_i = 0$	(off)	✓	23

Study A compares Cases 1 and 2, Study B Cases 2 and 3, Study C Cases 1 and 4. Primary control (governor droop) is active in every case. The indicators γ_{agc} (AGC present) and γ_g (GFM present) used below are therefore not switches inside one model: they mark which equations exist in each case (Case 1: $\gamma_{\text{agc}} = \gamma_g = 0$; Case 2: $\gamma_{\text{agc}} = 1$, $\gamma_g = 0$; Case 3: $\gamma_{\text{agc}} = \gamma_g = 1$; Case 4: $\gamma_g = 1$ with $K_i = 0$). Bus 5 (the GFM connection point) and its transformer x_{T5} exist only in the GFM model; Cases 1 and 2 use the 4-bus network of buses 1–4 (Section 2.4).

1 Base quantities, frames and per-unit conventions

Per-unit system and phasor representation: [3, Ch. 3], [4, Ch. 3].

1.1 Bases

$$S_{\text{base}} = 100 \text{ MVA}, \qquad f_0 = 50 \text{ Hz}, \qquad \omega_s = 2\pi f_0 \text{ rad/s}, \tag{1.1}$$

$$k_{S1} = \frac{S_1}{S_{\text{base}}}, \qquad k_{S2} = \frac{S_2}{S_{\text{base}}}, \qquad k_{Sg} = \frac{S_g}{S_{\text{base}}}. \tag{1.2}$$

1.2 Change of power base

A power P [MW] expressed on two bases:

$$P = p^{(100)}\, S_{\text{base}} = p^{(\text{own})}\, S_i \quad \Longrightarrow \quad p^{(\text{own})} = \frac{p^{(100)}}{k_{Si}}. \tag{1.3}$$

This is the gain to own base $= 1/k_{Si}$ used in every source block.

1.3 Change of impedance base

With equal voltage bases, $Z_{\text{base}} = V_{\text{base}}^2/S$, so

$$x^{(100)} = x^{(\text{own})}\,\frac{S_{\text{base}}}{S_i} \quad\Longrightarrow\quad x_{q1} = \frac{x_q}{k_{S1}}, \qquad x_{q2} = \frac{x_q}{k_{S2}}, \qquad x_g = \frac{0.15}{k_{Sg}}. \tag{1.4}$$

Current base $I_b = S/V_b$, so a current changes base inversely to the impedance:

$$i^{(\text{own})} = \frac{i^{(100)}}{k_{Si}} \qquad (\text{gain Id to own base} = 1/k_{Si}), \qquad x^{(100)} i^{(100)} = x^{(\text{own})} i^{(\text{own})}. \tag{1.5}$$

1.4 Phasor (RMS) frame

Every voltage is written as

$$v(t) = \sqrt{2}\,|V|\cos\!\big(\omega_s t + \theta(t)\big) \quad\longleftrightarrow\quad V = |V|e^{j\theta}, \tag{1.6}$$

i.e. all angles $(\delta_1, \delta_2, \delta_g, \theta_k)$ are measured in a reference frame rotating at the synchronous speed ω_s. The instantaneous angular frequency of a phasor is

$$\omega(t) = \frac{\mathrm{d}}{\mathrm{d}t}\big(\omega_s t + \theta\big) = \omega_s + \dot\theta \quad\Longrightarrow\quad \omega^{\text{pu}} = 1 + \frac{\dot\theta}{\omega_s}, \qquad \dot\theta = \omega_s\big(\omega^{\text{pu}} - 1\big). \tag{1.7}$$

Equation (1.7) is used three times in the model: for the rotor angle of each SG, for the GFM angle, and (inverted) for the bus-5 frequency measurement.

1.5 Complex power

For a current I leaving a node of voltage V,

$$S = P + jQ = V\,\overline{I}. \tag{1.8}$$

2 Model workspace: network data, load flow and initialisation

2.1 Branch data

Branch	x [pu, 100 MVA]	Element
1–2	$x_{T1} = 0.01$	transformer of SG1
5–2	$x_{T5} = 0.01$	transformer of GFM (Cases 3, 4 only)
3–4	$x_{T2} = 0.003$	transformer of SG2
2–3	$x_L = 0.2/2 = 0.1$	two parallel tie-lines

Two equal lines in parallel: $y_L = \frac{1}{j0.2} + \frac{1}{j0.2} = \frac{1}{j0.1}$, hence $x_L = 0.2/2$. The equations below are written for the 5-bus network of the GFM model; the 4-bus network of Cases 1 and 2 is given in Section 2.4.

2.2 Bus admittance matrix

Formulation: [10]. Kirchhoff's current law at bus i with a series branch i–j of admittance $y_{ij} = 1/(jx_{ij})$:

$$I_i = \sum_j y_{ij}\,(V_i - V_j) = \Big(\sum_j y_{ij}\Big)V_i - \sum_j y_{ij} V_j. \tag{2.1}$$

Each branch therefore adds the stamp

$$\begin{bmatrix} Y_{ii} & Y_{ij} \\ Y_{ji} & Y_{jj} \end{bmatrix} \mathrel{+}= \begin{bmatrix} y_{ij} & -y_{ij} \\ -y_{ij} & y_{ij} \end{bmatrix}, \qquad \boldsymbol{I} = \boldsymbol{Y}_{\text{bus}}\,\boldsymbol{V}, \quad \boldsymbol{Y}_{\text{bus}} \in \mathbb{C}^{5\times 5}. \tag{2.2}$$

2.3 Load-flow equations

Newton–Raphson power flow: [9], [10], [11]. Injected complex power at every bus:

$$S_i = V_i\,\overline{I_i} = V_i\,\overline{\textstyle\sum_{k=1}^{5} Y_{ik}V_k}, \qquad \boldsymbol{S} = \boldsymbol{V}\circ\overline{\boldsymbol{Y}_{\text{bus}}\boldsymbol{V}}. \tag{2.3}$$

In polar form, with $Y_{ik} = G_{ik} + jB_{ik}$ and $\theta_{ik} = \theta_i - \theta_k$:

$$P_i = \sum_k |V_i||V_k|\big(G_{ik}\cos\theta_{ik} + B_{ik}\sin\theta_{ik}\big), \tag{2.4}$$

$$Q_i = \sum_k |V_i||V_k|\big(G_{ik}\sin\theta_{ik} - B_{ik}\cos\theta_{ik}\big). \tag{2.5}$$

Bus types: bus 4 slack ($|V_4| = 1,\ \theta_4 = 0$), bus 1 PV ($P_1 = P_{SG1},\ |V_1| = 1$), buses 2, 3, 5 PQ. Unknowns and specified values:

$$\boldsymbol{z} = [\theta_1,\ \theta_2,\ \theta_3,\ \theta_5,\ |V_2|,\ |V_3|,\ |V_5|]^{\mathsf{T}}, \tag{2.6}$$

$$\boldsymbol{s} = \frac{1}{S_{\text{base}}}\,[P_{SG1},\ -P_{L2},\ -P_{L3},\ P_{GFM},\ 0,\ 0,\ Q_{GFM}]^{\mathsf{T}}, \tag{2.7}$$

$$\boldsymbol{g}(\boldsymbol{z}) = \big[P_1,\,P_2,\,P_3,\,P_5,\,Q_2,\,Q_3,\,Q_5\big]^{\mathsf{T}}(\boldsymbol{z}) - \boldsymbol{s} = \boldsymbol{0}. \tag{2.8}$$

Newton–Raphson: first-order Taylor expansion $\boldsymbol{g}(\boldsymbol{z}+\Delta\boldsymbol{z}) \approx \boldsymbol{g}(\boldsymbol{z}) + \boldsymbol{J}\Delta\boldsymbol{z} = \boldsymbol{0}$ gives

$$\boldsymbol{z}^{(n+1)} = \boldsymbol{z}^{(n)} - \boldsymbol{J}^{-1}\boldsymbol{g}(\boldsymbol{z}^{(n)}), \qquad J_{:,k} \approx \frac{\boldsymbol{g}(\boldsymbol{z}+h\boldsymbol{e}_k) - \boldsymbol{g}(\boldsymbol{z})}{h}, \quad h = 10^{-7}. \tag{2.9}$$

Result: $\boldsymbol{V}^0$ (bus voltages) and $\boldsymbol{S}^0 = \boldsymbol{V}^0 \circ \overline{\boldsymbol{Y}_{\text{bus}}\boldsymbol{V}^0}$ (bus injections).

2.4 Network of the cases without the GFM (Cases 1 and 2)

Without the GFM, bus 5 would be an open-ended bus behind x_{T5} with no injection, so its branch would carry no current and $V_5 = V_2$; it is therefore removed and the network has buses 1–4 only:

$$\text{branches } \{1\text{–}2,\ 3\text{–}4,\ 2\text{–}3\}, \qquad \boldsymbol{Y}_{\text{bus}} \in \mathbb{C}^{4\times 4}, \tag{2.10}$$

$$\boldsymbol{z} = [\theta_1,\ \theta_2,\ \theta_3,\ |V_2|,\ |V_3|]^{\mathsf{T}}, \qquad \boldsymbol{s} = \frac{1}{S_{\text{base}}}\,[P_{SG1},\ -P_{L2},\ -P_{L3},\ 0,\ 0]^{\mathsf{T}}, \tag{2.11}$$

$$\boldsymbol{g}(\boldsymbol{z}) = \big[P_1,\,P_2,\,P_3,\,Q_2,\,Q_3\big]^{\mathsf{T}}(\boldsymbol{z}) - \boldsymbol{s} = \boldsymbol{0}, \tag{2.12}$$

with bus 4 slack and bus 1 PV as before. Removing the current-free bus 5 does not change the operating point or the dynamic response of buses 1–4.

2.5 Machine (d,q) axes

The q-axis of a machine lies along the rotor angle δ, the d-axis lags it by 90° ($e^{j(\delta-\pi/2)}$). Projection of any network phasor X on these axes:

$$X = (X_d + jX_q)\,e^{j(\delta-\pi/2)} \implies X_d + jX_q = X\,e^{-j(\delta-\pi/2)} = jX\,e^{-j\delta}. \tag{2.13}$$

2.6 Stator equations of model 1.0

Park's transformation [5]; one-axis (flux-decay) model [4], [3]; the designation "model 1.0" (one d-axis and no q-axis rotor circuit) follows [13]. The full machine data are those of the GENROU

model in [1], [6]. Park's stator equations with $R_s = 0$, stator transients neglected and $\omega \approx 1$ in the EMF terms:

$$V_d = -\psi_q, \qquad V_q = \psi_d. \tag{2.14}$$

q-axis: no rotor circuit in model 1.0, so $\psi_q = -x_q I_q$. d-axis: field winding only,

$$\psi_d = -x_d I_d + x_{md} i_{fd}, \qquad \psi_{fd} = -x_{md} I_d + x_{fd} i_{fd}. \tag{2.15}$$

Eliminating i_{fd} with $i_{fd} = (\psi_{fd} + x_{md} I_d)/x_{fd}$:

$$\psi_d = -\Big(x_d - \frac{x_{md}^2}{x_{fd}}\Big) I_d + \frac{x_{md}}{x_{fd}} \psi_{fd} = -x'_d I_d + E'_q, \qquad x'_d \triangleq x_d - \frac{x_{md}^2}{x_{fd}}, \quad E'_q \triangleq \frac{x_{md}}{x_{fd}} \psi_{fd}. \tag{2.16}$$

Hence

$$\boxed{V_d = x_q I_q, \qquad V_q = E'_q - x'_d I_d.} \tag{2.17}$$

2.7 Voltage behind x_q (E_Q)

Define the phasor $\bar{E}_Q = V + j x_q I$. In (d, q) coordinates, using (2.13):

$$j \bar{E}_Q e^{-j\delta} = (V_d + j V_q) + j x_q (I_d + j I_q) = (V_d - x_q I_q) + j (V_q + x_q I_d). \tag{2.18}$$

By (2.17) the real part is zero, so $\bar{E}_Q$ lies exactly on the q-axis:

$$\boxed{\bar{E}_Q = E_Q \, e^{j\delta}, \qquad E_Q = V_q + x_q I_d = E'_q + (x_q - x'_d)\, I_d.} \tag{2.19}$$

This is why the network can treat each SG as a (time-varying) voltage $E_Q e^{j\delta}$ behind the constant reactance x_q.

2.8 Initial values of the synchronous generators

From the load-flow solution:

$$I_1 = \overline{(S_1^0 / V_1^0)}, \qquad \bar{E}_{Q1} = V_1^0 + j x_{q1} I_1, \qquad \delta_1^0 = \arg \bar{E}_{Q1}, \tag{2.20}$$

$$I_2 = \overline{(S_4^0 / V_4^0)}, \qquad \bar{E}_{Q2} = V_4^0 + j x_{q2} I_2, \qquad \delta_2^0 = \arg \bar{E}_{Q2}. \tag{2.21}$$

d-axis current on own base (Section 2.5 with I/k_{Si}):

$$I_{di}^0 = \mathrm{Re}\left\{ j \, \frac{I_i}{k_{Si}} \, e^{-j\delta_i^0} \right\}. \tag{2.22}$$

Transient EMF from (2.19), field voltage from $\dot{E}'_q = 0$ in (4.4) (Section 4.1):

$$E'^0_{qi} = |\bar{E}_{Qi}| - (x_q - x'_d)\, I_{di}^0, \qquad E_{fdi}^0 = E'^0_{qi} + (x_d - x'_d)\, I_{di}^0. \tag{2.23}$$

Exciter reference from the steady state of the exciter, $E_{fd} = K_A (V_{\mathrm{ref}} - V_t)$:

$$V_{ti}^0 = |V_{k_i}^0|, \qquad V_{\mathrm{ref},i} = V_{ti}^0 + \frac{E_{fdi}^0}{K_A}. \tag{2.24}$$

2.9 Initial value of the GFM

$$E_g = V_5^0 + j x_g \, \overline{(S_5^0 / V_5^0)}, \qquad |E_g| = E_{g,\mathrm{mag}} \text{ (held fixed)}, \qquad \delta_g^0 = \arg E_g. \tag{2.25}$$

Mechanical, governor and GFM set-points on own base:

$$P_{m1}^0 = \frac{\mathrm{Re}\, S_1^0}{k_{S1}}, \qquad P_{m2}^0 = \frac{\mathrm{Re}\, S_4^0}{k_{S2}}, \qquad P_{g,\mathrm{ref}} = \frac{\mathrm{Re}\, S_5^0}{k_{Sg}}, \qquad \theta_5^0 = \arg V_5^0, \tag{2.26}$$

$$P_{\mathrm{tie}}^0 = \mathrm{Re}\left\{ V_2^0 \, \overline{\left(\frac{V_2^0 - V_3^0}{j x_L} \right)} \right\}. \tag{2.27}$$

2.10 Constant-impedance loads

A load consuming S_L at voltage V represented by a shunt admittance y_L:

$$S_L = V\,\overline{(y_L V)} = \overline{y_L}\,|V|^2 \quad\Longrightarrow\quad y_L = \frac{\overline{S_L}}{|V|^2}. \tag{2.28}$$

Since the injection at a load bus is $S_i^0 = -S_{L,i}$:

$$y_{L2} = \frac{\overline{(-S_2^0)}}{|V_2^0|^2}, \qquad y_{L3} = \frac{\overline{(-S_3^0)}}{|V_3^0|^2}. \tag{2.29}$$

Consequence: the actual load power is voltage dependent, $P_{Li}(t) = G_{Li}\,|V_i(t)|^2$.

2.11 Norton equivalent of the sources (SGs and GFM)

Every source is a voltage behind a reactance: SGi is $\bar{E}_{Qi}$ behind jx_{qi} at bus k_i ($k_1 = 1$, $k_2 = 4$), the GFM is E_g behind jx_g at bus 5. For a source E behind jx connected to bus k:

$$I_{\text{out}} = \frac{E - V_k}{jx} = \underbrace{\frac{E}{jx}}_{\text{current source } J} - \underbrace{\frac{1}{jx}}_{\text{shunt}} V_k. \tag{2.30}$$

The shunt part is moved into the admittance matrix, the current source becomes the injection. The GFM is treated exactly like the two SGs:

$$J_1 = \frac{\bar{E}_{Q1}}{jx_{q1}}, \qquad J_2 = \frac{\bar{E}_{Q2}}{jx_{q2}}, \qquad J_g = \frac{E_g}{jx_g}. \tag{2.31}$$

2.12 Augmented admittance matrix and impedance matrix

$$\boldsymbol{Y}_{\text{aug}} = \boldsymbol{Y}_{\text{bus}} + \text{diag}(0,\, y_{L2},\, y_{L3},\, 0,\, 0), \tag{2.32}$$

$$Y_{\text{aug},11} \mathrel{+}= \frac{1}{jx_{q1}}, \qquad Y_{\text{aug},44} \mathrel{+}= \frac{1}{jx_{q2}}, \qquad Y_{\text{aug},55} \mathrel{+}= \frac{\gamma_g}{jx_g}. \tag{2.33}$$

Nodal equation with current-source injections only at buses 1, 4, 5:

$$\boldsymbol{Y}_{\text{aug}}\boldsymbol{V} = \boldsymbol{e}_1 J_1 + \boldsymbol{e}_4 J_2 + \boldsymbol{e}_5 J_g \quad\Longrightarrow\quad \boldsymbol{V} = \boldsymbol{Z}\,(\boldsymbol{e}_1 J_1 + \boldsymbol{e}_4 J_2 + \boldsymbol{e}_5 J_g), \qquad \boldsymbol{Z} = \boldsymbol{Y}_{\text{aug}}^{-1}. \tag{2.34}$$

Keeping only the columns of the injected buses:

$$\boxed{\boldsymbol{V} = \boldsymbol{Z}_n \begin{bmatrix} J_1 \\ J_2 \\ J_g \end{bmatrix}, \qquad \boldsymbol{Z}_n = \boldsymbol{Z}_{:,[1,4,5]} \in \mathbb{C}^{5\times 3} \quad \text{(Case 3)}.} \tag{2.35}$$

In Cases 1 and 2 (4-bus network, no GFM): $\boldsymbol{Y}_{\text{aug}} = \boldsymbol{Y}_{\text{bus}} + \text{diag}(0, y_{L2}, y_{L3}, 0)$ with the two SG shunts, $\boldsymbol{Z}_n = \boldsymbol{Z}_{:,[1,4]} \in \mathbb{C}^{4\times 2}$ and $\boldsymbol{V} = \boldsymbol{Z}_n[J_1;\, J_2]$.

2.13 Load step as a change of the impedance matrix

A step ΔP_{L2} (`dP_L2_MW`) at bus 2 is a conductance change at nominal voltage:

$$\Delta G_{L2} = \frac{\Delta P_{L2}/S_{\text{base}}}{|V_2^0|^2}, \qquad Y_{\text{aug},22}^{\text{post}} = Y_{\text{aug},22}^{\text{pre}} + \Delta G_{L2}, \tag{2.36}$$

$$\boldsymbol{Z}^{\text{post}} = \left(\boldsymbol{Y}_{\text{aug}}^{\text{post}}\right)^{-1}, \qquad \boldsymbol{Z}_n^{\text{post}} = \boldsymbol{Z}_{:,[1,4,5]}^{\text{post}}. \tag{2.37}$$

With $\Delta P_{L2} = -100$ MW this is a load rejection in Area 1.

2.14 Consistency check at $t = 0$

The model checks

$$\boldsymbol{Z}_n^{\text{pre}} \begin{bmatrix} \bar{E}_{Q1}/(jx_{q1}) \\ \bar{E}_{Q2}/(jx_{q2}) \\ E_g/(jx_g) \end{bmatrix} = \boldsymbol{V}^0. \tag{2.38}$$

At the operating point $P_{GFM} = Q_{GFM} = 0$, so from the GFM initialisation $E_g = V_5^0$. The Norton source current and the current drawn by the shunt $1/(jx_g)$ then cancel,

$$I_g^0 = J_g^0 - \frac{V_5^0}{jx_g} = \frac{E_g - V_5^0}{jx_g} = 0, \tag{2.39}$$

so adding the GFM does not disturb the load flow: its net contribution at $t = 0$ is zero.

3 Network block (algebraic equations)

Inputs: $\delta_1,\ \delta_2,\ \delta_g,\ u_{\text{step}},\ E_{Q1},\ E_{Q2}$. Outputs: $P_{e1}, P_{e2}, P_g, Q_g, \theta_5, |\boldsymbol{V}|, P_{\text{tie}}, Q_{e1}, Q_{e2}, I_{d1}, I_{d2}, V_{t1}, V_{t2}$.

3.1 Source voltages (**Source voltages**)

SG voltages from (2.19): magnitude E_{Qi} comes from the field-winding block (time varying), angle from the swing equation. GFM: fixed magnitude, controlled angle.

$$E_1 = E_{Q1}\, e^{j\delta_1}, \qquad E_2 = E_{Q2}\, e^{j\delta_2}, \qquad E_g = |E_g|\, e^{j\delta_g}. \tag{3.1}$$

3.2 Network solution (**Network solution**)

Step 1: impedance selection by the load step. The step signal $u_{\text{step}}(t) = 0$ for $t < t_{\text{step}}$, 1 for $t \geq t_{\text{step}}$ drives a switch (Select Zn, threshold 0.5):

$$\boldsymbol{Z}_n(t) = \begin{cases} \boldsymbol{Z}_n^{\text{pre}}, & u_{\text{step}} < 0.5 \\ \boldsymbol{Z}_n^{\text{post}}, & u_{\text{step}} \geq 0.5 \end{cases}. \tag{3.2}$$

Step 2: Norton currents of the sources. (Blocks Norton SG1 1 over jxq1, Norton SG2 1 over jxq2, Norton GFM 1 over jxg.)

$$J_1 = \frac{E_1}{jx_{q1}}, \qquad J_2 = \frac{E_2}{jx_{q2}}, \qquad J_g = \frac{E_g}{jx_g}. \tag{3.3}$$

Step 3: all bus voltages.

$$\boxed{\boldsymbol{V} = \boldsymbol{Z}_n \begin{bmatrix} J_1 \\ J_2 \\ J_g \end{bmatrix} = \boldsymbol{Z}_{:,1}\frac{E_1}{jx_{q1}} + \boldsymbol{Z}_{:,4}\frac{E_2}{jx_{q2}} + \boldsymbol{Z}_{:,5}\frac{E_g}{jx_g}.} \tag{3.4}$$

Step 4: GFM output current. The current that the GFM delivers to bus 5 is the Norton source current minus the current through its own shunt (blocks V bus 5, Eg - V5, GFM current 1 over jxg):

$$\boxed{I_g = J_g - \frac{V_5}{jx_g} = \frac{E_g - V_5}{jx_g}.} \tag{3.5}$$

Equivalence with the Thevenin form. Let $\boldsymbol{Y}_0 = \boldsymbol{Y}_{\text{aug}} - \frac{1}{jx_g}\boldsymbol{e}_5\boldsymbol{e}_5^{\mathsf{T}}$ be the network without the GFM, $\boldsymbol{Z}^{(0)} = \boldsymbol{Y}_0^{-1}$. Substituting $J_g = I_g + V_5/(jx_g)$ in the nodal equation gives $\boldsymbol{Y}_0\boldsymbol{V} = \boldsymbol{e}_1 J_1 + \boldsymbol{e}_4 J_2 + \boldsymbol{e}_5 I_g$, hence $V_5 = V_{th} + Z_{th} I_g$ with $V_{th} = Z_{51}^{(0)} J_1 + Z_{54}^{(0)} J_2$, $Z_{th} = Z_{55}^{(0)}$, and with $I_g = (E_g - V_5)/(jx_g)$:

$$I_g = \frac{E_g - V_{th}}{jx_g + Z_{th}}. \tag{3.6}$$

The Norton form used in the model and this Thevenin form give identical $\boldsymbol{V}$ and I_g; the Norton form treats the GFM in the same way as the SGs.

3.3 Source powers (Source powers)

Terminal current, power and d-axis current of SGi (terminal bus $k_1 = 1$, $k_2 = 4$):

$$I_{t1} = \frac{E_1 - V_1}{jx_{q1}}, \qquad S_{e1} = V_1\,\overline{I_{t1}}, \qquad P_{e1} = \operatorname{Re} S_{e1}, \qquad Q_{e1} = \operatorname{Im} S_{e1}, \tag{3.7}$$

$$I_{t2} = \frac{E_2 - V_4}{jx_{q2}}, \qquad S_{e2} = V_4\,\overline{I_{t2}}, \qquad P_{e2} = \operatorname{Re} S_{e2}, \qquad Q_{e2} = \operatorname{Im} S_{e2}, \tag{3.8}$$

$$S_g = V_5\,\overline{I_g}, \qquad P_g = \operatorname{Re} S_g, \qquad Q_g = \operatorname{Im} S_g. \tag{3.9}$$

Rotation to machine axes. From (2.13), on 100 MVA:

$$\boxed{I_{di} + jI_{qi} = j\,I_{ti}\,e^{-j\delta_i}, \qquad I_{di} = \operatorname{Re}\{jI_{ti}e^{-j\delta_i}\}.} \tag{3.10}$$

(Blocks: exp(-j delta) → Rotate to machine axes → times j → Id = Re.)

Terminal voltages for the exciters.

$$V_{t1} = |V_1|, \qquad V_{t2} = |V_4|. \tag{3.11}$$

Terminal power equals power at $\bar{E}_Q$ (lossless reactance).

$$E\overline{I} - V\overline{I} = (E-V)\overline{I} = jx_q\,I\overline{I} = jx_q|I|^2 \implies \operatorname{Re}\{E\overline{I}\} = \operatorname{Re}\{V\overline{I}\}, \qquad \operatorname{Im}\{E\overline{I}\} = \operatorname{Im}\{V\overline{I}\} + x_q|I|^2. \tag{3.12}$$

Hence P_{e1}, P_{e2}, P_g computed at the terminals are the electrical powers that enter the swing equation and the GFM droop.

Explicit nonlinear form. For $\bar{E}_Q = E_Q e^{j\delta}$, $V = |V|e^{j\theta}$:

$$S = V\,\overline{\frac{\bar{E}_Q - V}{jx_q}} = \frac{j\big(|V|E_Q e^{-j(\delta-\theta)} - |V|^2\big)}{x_q}, \tag{3.13}$$

$$\boxed{P_e = \frac{E_Q|V|}{x_q}\sin(\delta-\theta), \qquad Q_e = \frac{E_Q|V|\cos(\delta-\theta) - |V|^2}{x_q}.} \tag{3.14}$$

Same power in terms of E'_q (transient saliency). With $V_d = |V|\sin(\delta-\theta)$, $V_q = |V|\cos(\delta-\theta)$ and (2.17): $I_q = V_d/x_q$, $I_d = (E'_q - V_q)/x'_d$, so (own base)

$$P_e = V_d I_d + V_q I_q = \frac{E'_q|V|}{x'_d}\sin(\delta-\theta) + \frac{|V|^2}{2}\Big(\frac{1}{x_q} - \frac{1}{x'_d}\Big)\sin 2(\delta-\theta). \tag{3.15}$$

Substituting $E_Q = E'_q + (x_q - x'_d)I_d$ in the boxed P_e gives the same result, i.e. the E_Q-behind-x_q network model is exact for model 1.0.

3.4 Bus quantities (**Bus quantities**)

Voltage magnitudes. $|V_k|$, $k = 1, \ldots, 5$ ($k = 1, \ldots, 4$ in Cases 1 and 2).

Tie-line power (both lines, bus 2 → 3).

$$I_{\text{line}} = \frac{V_2 - V_3}{jx_L}, \qquad S_{\text{tie}} = V_2\,\overline{I_{\text{line}}}, \qquad \boxed{P_{\text{tie}} = \operatorname{Re} S_{\text{tie}} = \frac{|V_2||V_3|}{x_L}\sin(\theta_2 - \theta_3).} \tag{3.16}$$

(Same steps as the two-bus element above with $E \to V_2$, $V \to V_3$ and $\operatorname{Re}\{V_2\overline{I}\} = \operatorname{Re}\{V_3\overline{I}\}$ for a lossless line.)

Continuous (unwrapped) angle of bus 5. $\arg(\cdot)$ is limited to $(-\pi, \pi]$; to obtain a continuous angle the voltage is first rotated into the SG1 rotor frame:

$$\boxed{\theta_5 = \delta_1 + \arg\bigl(V_5\, e^{-j\delta_1}\bigr).} \tag{3.17}$$

Since $|\theta_5 - \delta_1| < \pi$ always holds, θ_5 is continuous even when δ_1 drifts by several turns.

3.5 Closed nonlinear form of the network (Kron-reduced)

Network reduction to internal nodes: [10, 4]. Combining the above, all source currents are linear in the internal voltages,

$$\begin{bmatrix} I_{t1} \\ I_{t2} \\ I_g \end{bmatrix} = \boldsymbol{Y}_{\text{red}} \begin{bmatrix} E_1 \\ E_2 \\ E_g \end{bmatrix}, \qquad \boldsymbol{Y}_{\text{red}} = \boldsymbol{Y}_d - \boldsymbol{Y}_d\,\tilde{\boldsymbol{Z}}_{[1,4,5],[1,4,5]}\,\boldsymbol{Y}_d, \tag{3.18}$$

$$\boldsymbol{Y}_d = \operatorname{diag}\Bigl(\tfrac{1}{jx_{q1}}, \tfrac{1}{jx_{q2}}, \tfrac{1}{jx_g}\Bigr), \qquad \tilde{\boldsymbol{Z}} = \boldsymbol{Z} = \boldsymbol{Y}_{\text{aug}}^{-1} \quad (x_g \text{ already in } \boldsymbol{Y}_{\text{aug}}). \tag{3.19}$$

Derivation: $I_{t,i} = (E_i - V_{k_i})/(jx_i) = Y_{d,ii}E_i - Y_{d,ii}V_{k_i}$ and $V_{k_i} = \sum_m \tilde{Z}_{k_i k_m} Y_{d,mm} E_m$ (Step 3 above). With $Y_{\text{red},ik} = G_{ik} + jB_{ik}$ and $\delta_{ik} = \delta_i - \delta_k$:

$$\boxed{P_i(\boldsymbol{\delta}, \boldsymbol{E}_Q) = \operatorname{Re}\{E_i\overline{I_i}\} = \sum_{k\in\{1,2,g\}} |E_i||E_k|\bigl(G_{ik}\cos\delta_{ik} + B_{ik}\sin\delta_{ik}\bigr)} \tag{3.20}$$

with $|E_1| = E_{Q1}(t)$, $|E_2| = E_{Q2}(t)$, $|E_g|$ constant. This is the nonlinear coupling of the angle states and of the field states (through E_{Qi}) through the network; $G_{ik} \neq 0$ because the constant-impedance loads are absorbed in $\boldsymbol{Y}_{\text{red}}$. The matrices change at t_{step} ($\boldsymbol{Y}_{\text{red}}^{\text{pre}} \to \boldsymbol{Y}_{\text{red}}^{\text{post}}$).

4 Synchronous generator blocks **SG1** and **SG2**

Both blocks are identical in structure; index $i \in \{1, 2\}$, rating S_i, k_{Si}, δ_i^0, P_{mi}^0, $E_{qi}'^0$, $V_{\text{ref},i}$. Inputs: P_{ei} [pu, 100 MVA], $\Delta P_{\text{ref},i}$ [pu own], I_{di} [pu, 100 MVA], V_{ti} [pu]. Outputs: δ_i, ω_i, P_{mi}, E_{Qi}.

4.1 Field winding, model 1.0 (**Field winding (model 1.0)**)

Derivation follows [4, Ch. 3], [3, Ch. 3]; model structure per [13].

Field circuit (Faraday + Ohm), per unit with time in seconds.

$$\frac{1}{\omega_b}\frac{\mathrm{d}\psi_{fd}}{\mathrm{d}t} = v_{fd} - r_{fd}\, i_{fd}. \tag{4.1}$$

Multiply by x_{md}/r_{fd} and use $E'_q = \frac{x_{md}}{x_{fd}}\psi_{fd}$:

$$\underbrace{\frac{x_{fd}}{\omega_b r_{fd}}}_{T'_{d0}}\frac{\mathrm{d}E'_q}{\mathrm{d}t} = \underbrace{\frac{x_{md}}{r_{fd}}v_{fd}}_{E_{fd}} - \underbrace{x_{md}\, i_{fd}}_{E_q}. \tag{4.2}$$

Open-circuit EMF E_q from $i_{fd} = (\psi_{fd} + x_{md}I_d)/x_{fd}$:

$$E_q = x_{md} i_{fd} = E'_q + \frac{x_{md}^2}{x_{fd}} I_d = E'_q + (x_d - x'_d) I_d. \tag{4.3}$$

Therefore, with $I_{di}^{\mathrm{own}} = I_{di}/k_{Si}$ (gain Id to own base):

$$\boxed{T'_{d0}\frac{\mathrm{d}E'_{qi}}{\mathrm{d}t} = E_{fdi} - E'_{qi} - (x_d - x'_d)\, I_{di}^{\mathrm{own}}, \qquad E'_{qi}(0) = E'^{\,0}_{qi},} \tag{4.4}$$

$$\boxed{E_{Qi} = E'_{qi} + (x_q - x'_d)\, I_{di}^{\mathrm{own}}} \qquad \text{(Voltage behind Xq, from (2.19)).} \tag{4.5}$$

(Blocks: Field balance → 1 over Td0p → Transient emf Eqp → Voltage behind Xq.)

4.2 Excitation system (Exciter (IEEE ST1))

Model ST1 of [7], see also [3, Ch. 8]; parameters from [1, Table 6].

Voltage transducer. First-order lag, state v_{Ri}:

$$\boxed{T_R\frac{\mathrm{d}v_{Ri}}{\mathrm{d}t} = V_{ti} - v_{Ri}, \qquad v_{Ri}(0) = V_{ti}^0,} \qquad \frac{V_{Ri}}{V_{ti}}(s) = \frac{1}{1+sT_R}. \tag{4.6}$$

Voltage error.

$$u_{Ei} = V_{\mathrm{ref},i} - v_{Ri}. \tag{4.7}$$

Transient gain reduction (lead–lag). Required: $\dfrac{Y}{U}(s) = \dfrac{1+sT_C}{1+sT_B}$. Split into a direct and a lag part:

$$\frac{1+sT_C}{1+sT_B} = \frac{T_C}{T_B} + \frac{1 - T_C/T_B}{1+sT_B} \quad \Big(\text{check: } \tfrac{T_C}{T_B}(1+sT_B) + 1 - \tfrac{T_C}{T_B} = 1 + sT_C\Big). \tag{4.8}$$

Realisation with one state x_{LLi}:

$$\boxed{T_B\frac{\mathrm{d}x_{LLi}}{\mathrm{d}t} = u_{Ei} - x_{LLi}, \qquad y_i = \frac{T_C}{T_B}u_{Ei} + \Big(1 - \frac{T_C}{T_B}\Big)x_{LLi}, \qquad x_{LLi}(0) = \frac{E^0_{fdi}}{K_A}.} \tag{4.9}$$

Exciter gain and ceiling.

$$\boxed{E_{fdi} = \mathrm{sat}_{[V_{R\min},\, V_{R\max}]}\,(K_A\, y_i).} \tag{4.10}$$

Overall (linear range).

$$E_{fdi}(s) = K_A \frac{1+sT_C}{1+sT_B}\Big(V_{\mathrm{ref},i} - \frac{V_{ti}(s)}{1+sT_R}\Big). \tag{4.11}$$

Steady state. $v_{Ri} = V_{ti}$, $x_{LLi} = u_{Ei}$, $y_i = u_{Ei}$, so $E_{fdi} = K_A(V_{\mathrm{ref},i} - V_{ti})$; with $V_{\mathrm{ref},i} = V_{ti}^0 + E_{fdi}^0/K_A$ the initial state is an equilibrium.

4.3 Swing equation (Swing equation)

Derivation follows [3, Ch. 3], [4, Ch. 3], [11].

Newton's second law for the rotor.

$$J\frac{\mathrm{d}\omega_m}{\mathrm{d}t} = T_m - T_e - T_D. \tag{4.12}$$

Inertia constant.

$$H = \frac{\frac{1}{2}J\omega_{sm}^2}{S_i}\ [\mathrm{s}] \quad\Longrightarrow\quad J = \frac{2HS_i}{\omega_{sm}^2}. \tag{4.13}$$

Per-unit form. Divide by the base torque $T_b = S_i/\omega_{sm}$ and write $\omega = \omega_m/\omega_{sm}$:

$$\frac{2HS_i}{\omega_{sm}^2}\omega_{sm}\frac{\mathrm{d}\omega}{\mathrm{d}t}\cdot\frac{\omega_{sm}}{S_i} = T_m^{\mathrm{pu}} - T_e^{\mathrm{pu}} - T_D^{\mathrm{pu}} \quad\Longrightarrow\quad 2H\frac{\mathrm{d}\omega}{\mathrm{d}t} = T_m^{\mathrm{pu}} - T_e^{\mathrm{pu}} - T_D^{\mathrm{pu}}. \tag{4.14}$$

Power to torque. $P = T\omega_m \Rightarrow T^{\mathrm{pu}} = P^{\mathrm{pu}}/\omega$. Damping $T_D^{\mathrm{pu}} = D(\omega - 1)$. Electrical power converted to own base, P_{ei}/k_{Si}:

$$\boxed{\frac{\mathrm{d}\omega_i}{\mathrm{d}t} = \frac{1}{2H}\left[\frac{P_{mi} - P_{ei}/k_{Si}}{\omega_i} - D(\omega_i - 1)\right], \qquad \omega_i(0) = 1.} \tag{4.15}$$

(Blocks: to own base → Accelerating power → Torque ($\div\omega$) → Minus damping → 1 over 2H → Speed.)

Rotor angle. Electrical rotor angle relative to the synchronous frame, from (1.7):

$$\boxed{\frac{\mathrm{d}\delta_i}{\mathrm{d}t} = \omega_s(\omega_i - 1), \qquad \delta_i(0) = \delta_i^0.} \tag{4.16}$$

(Blocks: Speed deviation → omega_s → Rotor angle.)

Second-order nonlinear form. Substituting (3.20) into (4.15)–(4.16):

$$\frac{2H}{\omega_s}\ddot{\delta}_i = \frac{P_{mi} - \frac{1}{k_{Si}}P_i(\delta_1, \delta_2, \delta_g, E_{Q1}, E_{Q2})}{1 + \dot{\delta}_i/\omega_s} - \frac{D}{\omega_s}\dot{\delta}_i. \tag{4.17}$$

Here $D = 2$ represents the damper windings that model 1.0 omits.

4.4 Speed governor (Governor (IEEEG1))

IEEEG1 governor [8, 12]; parameters from [1, Table 7]; see also [3, Ch. 9].

Load reference.

$$P_{\mathrm{ref},i} = P^0_{mi} + \Delta P_{\mathrm{ref},i} \quad (\Delta P_{\mathrm{ref},i} \text{ from the AGC}). \tag{4.18}$$

Governor error (droop feedback and valve-position feedback).

$$e_{gi} = P_{\mathrm{ref},i} - K(\omega_i - 1) - g_{vi}, \qquad K = \frac{1}{R}. \tag{4.19}$$

Servo motor with rate limit.

$$r_i = \mathrm{sat}_{[U_c,\,U_o]}\left(\frac{1}{T_3}\, e_{gi}\right), \tag{4.20}$$

Valve position with position limits (limited integrator).

$$\boxed{\frac{\mathrm{d}g_{vi}}{\mathrm{d}t} = \begin{cases} 0, & g_{vi} = P_{\max} \text{ and } r_i > 0, \\ 0, & g_{vi} = P_{\min} \text{ and } r_i < 0, \\ r_i, & \text{otherwise}, \end{cases} \qquad g_{vi}(0) = P^0_{mi}.} \tag{4.21}$$

Linear range. Without limits,

$$T_3 \dot{g}_{vi} = P_{\mathrm{ref},i} - K\Delta\omega_i - g_{vi} \implies G_{vi}(s) = \frac{1}{1+sT_3}\big(P_{\mathrm{ref},i}(s) - K\Delta\omega_i(s)\big). \tag{4.22}$$

Droop. In steady state $\dot{g}_{vi} = 0$:

$$g_{vi} = P_{\mathrm{ref},i} - K\Delta\omega_i \implies R = -\frac{\Delta\omega_i}{\Delta g_{vi}} = \frac{1}{K} = 0.05 \text{ pu } (5\,\%). \tag{4.23}$$

4.5 Reheat steam turbine (Turbine (reheat))

Tandem-compound single-reheat turbine [8, 12], [3, Ch. 9].

First-order lag of a steam volume. Mass in a volume m; outflow proportional to stored mass (pressure), $q_{\mathrm{out}} = m/T$:

$$\frac{\mathrm{d}m}{\mathrm{d}t} = q_{\mathrm{in}} - q_{\mathrm{out}} = q_{\mathrm{in}} - \frac{m}{T} \implies T\frac{\mathrm{d}q_{\mathrm{out}}}{\mathrm{d}t} = q_{\mathrm{in}} - q_{\mathrm{out}}, \qquad \frac{q_{\mathrm{out}}}{q_{\mathrm{in}}}(s) = \frac{1}{1+sT}. \tag{4.24}$$

Implemented in each sub-block as $\dot{x} = \frac{1}{T}(u - x)$ (Error → One over T → State).

Steam chest, reheater, crossover in cascade.

$$\frac{\mathrm{d}x_{HP,i}}{\mathrm{d}t} = \frac{1}{T_4}\big(g_{vi} - x_{HP,i}\big), \qquad x_{HP,i}(0) = P^0_{mi}, \tag{4.25}$$

$$\frac{\mathrm{d}x_{IP,i}}{\mathrm{d}t} = \frac{1}{T_5}\big(x_{HP,i} - x_{IP,i}\big), \qquad x_{IP,i}(0) = P^0_{mi}, \tag{4.26}$$

$$\frac{\mathrm{d}x_{LP,i}}{\mathrm{d}t} = \frac{1}{T_6}\big(x_{IP,i} - x_{LP,i}\big), \qquad x_{LP,i}(0) = P^0_{mi}. \tag{4.27}$$

Mechanical power.

$$\boxed{P_{mi} = K_1 x_{HP,i} + K_3 x_{IP,i} + K_5 x_{LP,i}, \qquad K_1 + K_3 + K_5 = 1.} \tag{4.28}$$

Transfer function.

$$\frac{P_{mi}}{G_{vi}}(s) = \frac{1}{1+sT_4}\left[K_1 + \frac{K_3}{1+sT_5} + \frac{K_5}{(1+sT_5)(1+sT_6)}\right], \qquad \left.\frac{P_{mi}}{G_{vi}}\right|_{s=0} = K_1 + K_3 + K_5 = 1. \tag{4.29}$$

4.6 Interconnection inside an SG block

$$\Delta P_{\mathrm{ref},i} \xrightarrow{\text{governor}} g_{vi} \xrightarrow{\text{turbine}} P_{mi} \xrightarrow{\text{swing}} (\omega_i, \delta_i), \qquad \omega_i \to \text{governor}, \quad P_{ei} \to \text{swing}, \tag{4.30}$$

$$V_{ti} \xrightarrow{\text{exciter}} E_{fdi} \xrightarrow{\text{field winding}} E'_{qi} \to E_{Qi} \to \text{network}, \qquad I_{di} \to \text{field winding}. \tag{4.31}$$

5 Grid-forming converter block GFM

GFM with P–f droop and POD-P stabilizer as proposed in [1, 2]. Inputs: P_g [pu, 100 MVA], f_{meas} [pu]. Outputs: δ_g, ω_g. The converter is an ideal voltage source $E_g = |E_g|e^{j\delta_g}$ (magnitude fixed) behind x_g; only its angle is controlled.

5.1 Power error and P–f droop

$$P_g^{\mathrm{own}} = \frac{P_g}{k_{Sg}}, \qquad \Delta\omega_{\mathrm{droop}} = R_g\big(P_{g,\mathrm{ref}} - P_g^{\mathrm{own}}\big). \tag{5.1}$$

Derivation of the droop law: a synchronous machine in steady state follows $\Delta\omega = -R\,\Delta P$; the GFM imposes the same relation directly on its internal frequency, so that P_g rises when frequency falls.

5.2 Power-oscillation damper (POD-P) with washout

Measured frequency deviation:

$$\Delta\omega_{\mathrm{meas}} = f_{\mathrm{meas}} - 1. \tag{5.2}$$

Washout realised with an integrator state z:

$$\boxed{\frac{\mathrm{d}z}{\mathrm{d}t} = \frac{\gamma_g}{T_w}\big(\Delta\omega_{\mathrm{meas}} - z\big), \qquad z(0) = 0,} \tag{5.3}$$

$$w = \Delta\omega_{\mathrm{meas}} - z, \qquad \frac{Z(s)}{\Delta\Omega_{\mathrm{meas}}(s)} = \frac{1}{1+sT_w} \implies \frac{W(s)}{\Delta\Omega_{\mathrm{meas}}(s)} = 1 - \frac{1}{1+sT_w} = \frac{sT_w}{1+sT_w}. \tag{5.4}$$

POD term: $u_{\mathrm{POD}} = K_w\,\gamma_{\mathrm{pod}}\,w$, $\gamma_{\mathrm{pod}} = \texttt{pod_on}$. The washout removes the steady-state component, so the POD acts only during transients.

5.3 Internal frequency

$$\boxed{\omega_g = 1 + R_g\big(P_{g,\mathrm{ref}} - P_g/k_{Sg}\big) - K_w\gamma_{\mathrm{pod}}\big(\Delta\omega_{\mathrm{meas}} - z\big).} \tag{5.5}$$

In Laplace form:

$$\Delta\Omega_g(s) = -R_g\,\Delta P_g^{\mathrm{own}}(s) - K_w\gamma_{\mathrm{pod}}\frac{sT_w}{1+sT_w}\,\Delta\Omega_{\mathrm{meas}}(s). \tag{5.6}$$

5.4 Converter angle

From (1.7):

$$\boxed{\frac{\mathrm{d}\delta_g}{\mathrm{d}t} = \gamma_g\,\omega_s(\omega_g - 1), \qquad \delta_g(0) = \delta_g^0.} \tag{5.7}$$

Combined first-order nonlinear form.

$$\frac{1}{\omega_s R_g}\,\dot{\delta}_g = P_{g,\mathrm{ref}} - \frac{1}{k_{Sg}} P_g(\delta_1, \delta_2, \delta_g) - \frac{K_w \gamma_{\mathrm{pod}}}{R_g}\big(\Delta\omega_{\mathrm{meas}} - z\big) \qquad (\gamma_g = 1). \tag{5.8}$$

Compared with the swing equation (second order, inertia $2H$), the droop GFM is first order: zero inertia and an equivalent damping coefficient $1/(\omega_s R_g)$.

Steady state. With $\dot{\delta}_g = 0$ in a common frequency ω and $w = 0$:

$$P_g^{\mathrm{own}} = P_{g,\mathrm{ref}} - \frac{\omega - 1}{R_g} \implies \Delta P_g^{(100)} = -\frac{k_{Sg}}{R_g}\Delta\omega. \tag{5.9}$$

6 Bus-5 frequency measurement (Bus 5 frequency measurement)

Ideal bus frequency from (1.7): $f_5 = 1 + \dot{\theta}_5/\omega_s$. A pure derivative is not realisable, so it is filtered:

$$\widehat{\dot{\theta}_5}(s) = \frac{s}{1 + sT_f}\,\theta_5(s). \tag{6.1}$$

Realisation with a state $\hat{\theta}_5$ (Filtered angle):

$$\boxed{\frac{\mathrm{d}\hat{\theta}_5}{\mathrm{d}t} = \frac{1}{T_f}\big(\theta_5 - \hat{\theta}_5\big), \qquad \hat{\theta}_5(0) = \theta_5^0,} \tag{6.2}$$

since $\hat{\theta}_5 = \frac{1}{1+sT_f}\theta_5 \Rightarrow s\hat{\theta}_5 = \frac{s}{1+sT_f}\theta_5$. Output:

$$\boxed{f_{\mathrm{meas}} = 1 + \frac{1}{\omega_s}\cdot\frac{\theta_5 - \hat{\theta}_5}{T_f}.} \tag{6.3}$$

$\hat{\theta}_5(0) = \theta_5^0$ gives $f_{\mathrm{meas}}(0) = 1$ (no start-up transient).

7 Secondary control block Secondary control (AGC)

Tie-line bias control: [3, Ch. 11], [11], [14]. Inputs (via Goto/From): ω_1, ω_2, P_{tie}. Outputs: $\Delta P_{\mathrm{ref},1}$, $\Delta P_{\mathrm{ref},2}$. Area 1 = SG1 (+ GFM), Area 2 = SG2.

7.1 Area frequency deviations and tie-line deviation

$$\Delta f_1 = \omega_1 - 1, \qquad \Delta f_2 = \omega_2 - 1, \qquad \Delta P_{\mathrm{tie}} = P_{\mathrm{tie}} - P_{\mathrm{tie}}^0. \tag{7.1}$$

For the lossless tie, the export of Area 2 is $\Delta P_{\mathrm{tie},2} = -\Delta P_{\mathrm{tie}}$.

7.2 Frequency bias = area frequency response characteristic

Static response of each area to a frequency change Δf (on 100 MVA): from the governor droop, $\Delta P_{mi}^{(100)} = -k_{Si} K \Delta f$; from (5.9), $\Delta P_g^{(100)} = -\frac{k_{Sg}}{R_g}\Delta f$. The loads are static (no load damping). The bias set in the AGC block uses the governor and GFM droops:

$$\boxed{B_1 = K\, k_{S1} + \gamma_g \frac{k_{Sg}}{R_g}, \qquad B_2 = K\, k_{S2} \qquad [\mathrm{pu}_{100}/\mathrm{pu}_f].} \tag{7.2}$$

The rotor damping D adds a further static response: from (4.15) at $\dot{\omega}_i = 0$, $P_{ei}/k_{Si} = P_{mi} - D\,\omega_i \Delta\omega_i \approx P_{mi} - D\Delta\omega_i$, i.e. $\Delta P_{ei}^{(100)} \approx \Delta P_{mi}^{(100)} - k_{Si} D \Delta f$. The true area frequency response characteristic is therefore

$$\beta_1 = (K + D)\, k_{S1} + \gamma_g \frac{k_{Sg}}{R_g}, \qquad \beta_2 = (K + D)\, k_{S2}, \tag{7.3}$$

and $B_i = \beta_i - D\, k_{Si}$ (bias slightly below β_i).

7.3 Area control errors

$$\boxed{ACE_1 = \Delta P_{\mathrm{tie}} + B_1 \Delta f_1, \qquad ACE_2 = -\Delta P_{\mathrm{tie}} + B_2 \Delta f_2.} \tag{7.4}$$

7.4 Integral control (set-point changes)

Converted to the own base of each machine:

$$\boxed{\frac{\mathrm{d}\,\Delta P_{\mathrm{ref},1}}{\mathrm{d}t} = -\gamma_{\mathrm{agc}} \frac{K_i}{k_{S1}}\, ACE_1, \qquad \frac{\mathrm{d}\,\Delta P_{\mathrm{ref},2}}{\mathrm{d}t} = -\gamma_{\mathrm{agc}} \frac{K_i}{k_{S2}}\, ACE_2,} \tag{7.5}$$

$\Delta P_{\mathrm{ref},i}(0) = 0$; $\gamma_{\mathrm{agc}} = 1$ in Cases 2 and 3 (in Case 4 the same block runs with $K_i = 0$), and the AGC block does not exist in Case 1.

Why integral control. In equilibrium the integrator inputs are zero, so $ACE_1 = ACE_2 = 0$; this is proved in Section 10.

8 Load step, scaling and logged outputs

$$u_{\mathrm{step}}(t) = \begin{cases} 0, & t < t_{\mathrm{step}} \\ 1, & t \geq t_{\mathrm{step}} \end{cases} \quad (\text{size } \Delta P_{L2} \text{ is inside } \boldsymbol{Z}_n^{\mathrm{post}}), \tag{8.1}$$

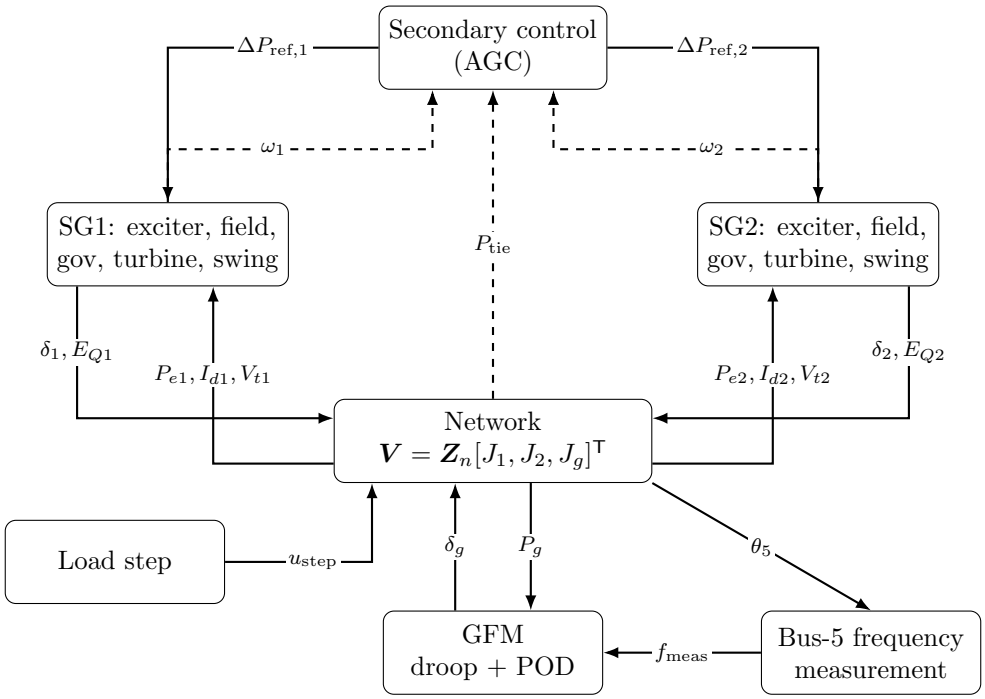
$$f_i\,[\mathrm{Hz}] = f_0\, \omega_i, \qquad f_5\,[\mathrm{Hz}] = f_0\, f_{\mathrm{meas}}, \tag{8.2}$$

$$P_{ei}\,[\mathrm{MW}] = S_{\mathrm{base}} P_{ei}, \quad P_g\,[\mathrm{MW}] = S_{\mathrm{base}} P_g, \quad P_{\mathrm{tie}}\,[\mathrm{MW}] = S_{\mathrm{base}} P_{\mathrm{tie}}, \tag{8.3}$$

$$P_{mi}\,[\mathrm{MW}] = S_i\, P_{mi}, \qquad Q\,[\mathrm{MVAr}] = S_{\mathrm{base}}\, Q. \tag{8.4}$$

9 Complete nonlinear model and interconnection

9.1 Signal flow of the top level



Dashed lines are Goto/From connections; I_{d1}, I_{d2}, V_{t1}, V_{t2} also reach the SG blocks through Goto/From. (δ_1, δ_2 are also used inside the network for I_{d1}, I_{d2} and δ_1 for the unwrapped angle θ_5.)

9.2 State vector (23 states in Case 3)

$$\boldsymbol{x}_{SGi} = \big[\delta_i,\ \omega_i,\ E'_{qi},\ v_{Ri},\ x_{LLi},\ g_{vi},\ x_{HP,i},\ x_{IP,i},\ x_{LP,i}\big]^{\mathsf{T}}, \qquad i = 1, 2, \tag{9.1}$$

$$\boldsymbol{x} = \big[\boldsymbol{x}_{SG1}^{\mathsf{T}},\ \boldsymbol{x}_{SG2}^{\mathsf{T}},\ \underbrace{\delta_g,\ z}_{\text{GFM}},\ \underbrace{\hat{\theta}_5}_{\text{meas.}},\ \underbrace{\Delta P_{\mathrm{ref},1},\ \Delta P_{\mathrm{ref},2}}_{\text{AGC}}\big]^{\mathsf{T}}. \tag{9.2}$$

Case 2 has no GFM and no bus-5 frequency measurement (20 states); Case 1 has in addition no AGC (18 states).

9.3 Differential–algebraic system

Simultaneous solution of the machine differential equations and the algebraic network equations, as in the transient-stability procedure of [11], extended to the multimachine model of [4].

Algebraic equations (network, solved at every instant):

$$E_{Qi} = E'_{qi} + (x_q - x'_d)\frac{I_{di}}{k_{Si}} \quad (i = 1, 2), \tag{9.3}$$

$$E_1 = E_{Q1}e^{j\delta_1}, \quad E_2 = E_{Q2}e^{j\delta_2}, \quad E_g = |E_g|e^{j\delta_g}, \tag{9.4}$$

$$\boldsymbol{V} = \boldsymbol{Z}_n\Big[\tfrac{E_1}{jx_{q1}},\ \tfrac{E_2}{jx_{q2}},\ \tfrac{E_g}{jx_g}\Big]^{\mathsf{T}}, \qquad I_g = \frac{E_g - V_5}{jx_g}, \tag{9.5}$$

$$I_{t1} = \tfrac{E_1 - V_1}{jx_{q1}}, \quad I_{t2} = \tfrac{E_2 - V_4}{jx_{q2}}, \quad I_{di} = \mathrm{Re}\{jI_{ti}e^{-j\delta_i}\}, \quad V_{t1} = |V_1|,\ V_{t2} = |V_4|, \tag{9.6}$$

$$P_{e1} = \mathrm{Re}\{V_1\overline{I_{t1}}\}, \quad P_{e2} = \mathrm{Re}\{V_4\overline{I_{t2}}\}, \quad P_g = \mathrm{Re}\{V_5\overline{I_g}\}, \tag{9.7}$$

$$P_{\mathrm{tie}} = \mathrm{Re}\Big\{V_2\overline{\tfrac{V_2 - V_3}{jx_L}}\Big\}, \qquad \theta_5 = \delta_1 + \arg(V_5 e^{-j\delta_1}), \tag{9.8}$$

$$f_{\mathrm{meas}} = 1 + \frac{\theta_5 - \hat{\theta}_5}{\omega_s T_f}, \qquad \omega_g = 1 + R_g\Big(P_{g,\mathrm{ref}} - \tfrac{P_g}{k_{Sg}}\Big) - K_w\gamma_{\mathrm{pod}}(f_{\mathrm{meas}} - 1 - z), \tag{9.9}$$

$$u_{Ei} = V_{\mathrm{ref},i} - v_{Ri}, \qquad E_{fdi} = \mathrm{sat}\Big(K_A\Big[\tfrac{T_C}{T_B}u_{Ei} + \big(1 - \tfrac{T_C}{T_B}\big)x_{LLi}\Big]\Big), \tag{9.10}$$

$$e_{gi} = P^0_{mi} + \Delta P_{\mathrm{ref},i} - K(\omega_i - 1) - g_{vi}, \qquad P_{mi} = K_1 x_{HP,i} + K_3 x_{IP,i} + K_5 x_{LP,i}. \tag{9.11}$$

Implicit loop $E_{Qi} \leftrightarrow I_{di}$. (9.3) needs I_{di}, which depends on E_{Q1}, E_{Q2} through the network. Because the network is linear in (E_{Q1}, E_{Q2}) for fixed angles,

$$I_{di} = \sum_{k=1,2} a_{ik}(\boldsymbol{\delta}, u_{\mathrm{step}})\, E_{Qk} + b_i(\boldsymbol{\delta}, u_{\mathrm{step}}), \tag{9.12}$$

so (9.3) is a 2×2 linear system for E_{Q1}, E_{Q2} at every instant; Simulink solves this algebraic loop numerically (`ode15s`, DAE index 1).

Differential equations ($i = 1, 2$):

$$\dot{\delta}_i = \omega_s(\omega_i - 1), \tag{9.13}$$

$$\dot{\omega}_i = \frac{1}{2H}\Big[\frac{P_{mi} - P_{ei}/k_{Si}}{\omega_i} - D(\omega_i - 1)\Big], \tag{9.14}$$

$$\dot{E}'_{qi} = \frac{1}{T'_{d0}}\Big[E_{fdi} - E'_{qi} - (x_d - x'_d)\frac{I_{di}}{k_{Si}}\Big], \tag{9.15}$$

$$\dot{v}_{Ri} = \frac{1}{T_R}(V_{ti} - v_{Ri}), \qquad \dot{x}_{LLi} = \frac{1}{T_B}(u_{Ei} - x_{LLi}), \tag{9.16}$$

$$\dot{g}_{vi} = \mathrm{sat}_{[U_c, U_o]}\Big(\frac{1}{T_3}e_{gi}\Big) \text{ with } g_{vi} \in [P_{\min}, P_{\max}], \tag{9.17}$$

$$\dot{x}_{HP,i} = (g_{vi} - x_{HP,i})/T_4, \quad \dot{x}_{IP,i} = (x_{HP,i} - x_{IP,i})/T_5, \quad \dot{x}_{LP,i} = (x_{IP,i} - x_{LP,i})/T_6, \tag{9.18}$$

$$\dot{\delta}_g = \gamma_g\,\omega_s(\omega_g - 1), \qquad \dot{z} = \frac{\gamma_g}{T_w}(f_{\mathrm{meas}} - 1 - z), \qquad \dot{\hat{\theta}}_5 = \frac{\theta_5 - \hat{\theta}_5}{T_f}, \tag{9.19}$$

$$\dot{\Delta P}_{\mathrm{ref},1} = -\gamma_{\mathrm{agc}}\frac{K_i}{k_{S1}}\big[\Delta P_{\mathrm{tie}} + B_1(\omega_1 - 1)\big], \qquad \dot{\Delta P}_{\mathrm{ref},2} = -\gamma_{\mathrm{agc}}\frac{K_i}{k_{S2}}\big[-\Delta P_{\mathrm{tie}} + B_2(\omega_2 - 1)\big]. \tag{9.20}$$

Solver: `ode15s`, RelTol $= 10^{-6}$, $\Delta t_{\max} = 0.01$ s.

9.4 Equilibrium at $t = 0$

Setting $\dot{\boldsymbol{x}} = \mathbf{0}$ with $\omega_i = 1$:

$$\dot{\omega}_i = 0 \Rightarrow P_{mi} = P^0_{ei}/k_{Si} = P^0_{mi}, \tag{9.21}$$

$$\dot{x}_{LP} = \dot{x}_{IP} = \dot{x}_{HP} = 0 \Rightarrow x_{LP,i} = x_{IP,i} = x_{HP,i} = g_{vi} = P^0_{mi}, \tag{9.22}$$

$$\dot{g}_{vi} = 0 \Rightarrow P_{\mathrm{ref},i} = g_{vi} = P^0_{mi} \Rightarrow \Delta P_{\mathrm{ref},i} = 0, \tag{9.23}$$

$$\dot{E}'_{qi} = 0 \Rightarrow E^0_{fdi} = E'^0_{qi} + (x_d - x'_d) I^0_{di}, \tag{9.24}$$

$$\dot{v}_{Ri} = \dot{x}_{LLi} = 0 \Rightarrow v_{Ri} = V^0_{ti},\ x_{LLi} = V_{\mathrm{ref},i} - V^0_{ti} = E^0_{fdi}/K_A, \tag{9.25}$$

$$\dot{\hat{\theta}}_5 = 0 \Rightarrow \hat{\theta}_5 = \theta^0_5 \Rightarrow f_{\mathrm{meas}} = 1, \tag{9.26}$$

$$\dot{z} = 0,\ \dot{\delta}_g = 0 \Rightarrow z = 0,\ P^{\mathrm{own}}_g = P_{g,\mathrm{ref}}, \tag{9.27}$$

$$\dot{\Delta P}_{\mathrm{ref},i} = 0 \Rightarrow P_{\mathrm{tie}} = P^0_{\mathrm{tie}}. \tag{9.28}$$

These are exactly the initial conditions placed in the integrators.

10 Steady-state response to the load step

Cf. the two-area analysis in [3, Ch. 11]. After the transient all units share one frequency, $\Delta\omega_1 = \Delta\omega_2 = \Delta\omega_g = \Delta f$. Let ΔP_L be the change of the Area-1 load (on 100 MVA).

10.1 Primary control only ($\gamma_{\mathbf{agc}} = 0$)

Electrical generation changes (governor droop + rotor damping + GFM droop): $\Delta P_{\mathrm{gen},1} = -\beta_1 \Delta f$, $\Delta P_{\mathrm{gen},2} = -\beta_2 \Delta f$. Power balance of each area (lossless network):

$$\text{Area 1:} \quad -\beta_1 \Delta f = \Delta P_L + \Delta P_{\mathrm{tie}}, \tag{10.1}$$

$$\text{Area 2:} \quad -\beta_2 \Delta f = -\Delta P_{\mathrm{tie}}. \tag{10.2}$$

Adding and solving:

$$\boxed{\Delta f = -\frac{\Delta P_L}{\beta_1 + \beta_2}, \qquad \Delta P_{\mathrm{tie}} = -\frac{\beta_2}{\beta_1 + \beta_2}\,\Delta P_L.} \tag{10.3}$$

(The actual ΔP_L is $\Delta G_{L2}|V_2|^2$, i.e. voltage dependent; the excitation systems keep $|V_2|$ close to its initial value.)

10.2 With supplementary control ($\gamma_{\mathbf{agc}} = 1$)

Equilibrium of the AGC integrators:

$$ACE_1 = \Delta P_{\mathrm{tie}} + B_1 \Delta f = 0, \qquad ACE_2 = -\Delta P_{\mathrm{tie}} + B_2 \Delta f = 0. \tag{10.4}$$

Adding: $(B_1 + B_2)\Delta f = 0$. Since $B_1 + B_2 > 0$ (the result does not depend on $B_i = \beta_i$):

$$\boxed{\Delta f = 0, \qquad \Delta P_{\mathrm{tie}} = 0, \qquad \Delta P_{\mathrm{gen},1} = \Delta P_L, \qquad \Delta P_{\mathrm{gen},2} = 0.} \tag{10.5}$$

Area 1 absorbs its own load change; with $\Delta f = 0$ the GFM returns to $P^{\mathrm{own}}_g = P_{g,\mathrm{ref}}$ by (5.9), so the whole change is finally taken by SG1.

11 Small-signal (linearised) form

Cf. [3, Ch. 12], [4].

11.1 Linearised network

From (3.20), around the operating point and for fixed E_{Q1}, E_{Q2}:

$$\Delta P_i = \sum_k \frac{\partial P_i}{\partial \delta_k}\Delta\delta_k, \qquad \frac{\partial P_i}{\partial \delta_k} = -|E_i||E_k|\big(G_{ik}\sin\delta_{ik} - B_{ik}\cos\delta_{ik}\big),\ k \neq i, \tag{11.1}$$

$$\frac{\partial P_i}{\partial \delta_i} = -\sum_{k\neq i}\frac{\partial P_i}{\partial \delta_k} \quad \text{(synchronising coefficients)}. \tag{11.2}$$

11.2 Linearised tie-line

$$\Delta P_{\text{tie}} \approx \frac{|V_2||V_3|}{x_L}\cos(\theta_2^0 - \theta_3^0)\,(\Delta\theta_2 - \Delta\theta_3) = T_{23}(\Delta\theta_2 - \Delta\theta_3), \tag{11.3}$$

and, using $\Delta heta_k(s) = \omega_s \Delta F_k(s)/s$,

$$\Delta P_{\text{tie}}(s) = \frac{\omega_s T_{23}}{s}\big(\Delta F_2(s) - \Delta F_3(s)\big), \qquad \Delta f_k = \frac{\dot\theta_k}{\omega_s}\ \text{(bus frequencies)}, \tag{11.4}$$

which is the classical two-area tie-line relation $\Delta P_{\text{tie}} = \frac{2\pi T_{12}}{s}(\Delta f_1 - \Delta f_2)$ (with f in Hz).

11.3 Linearised swing equation

With $\omega_i \approx 1$, $1/\omega_i \approx 1 - \Delta\omega_i$:

$$2H\,s\,\Delta\Omega_i = \Delta P_{mi} - \frac{\Delta P_{ei}}{k_{Si}} - D\Delta\Omega_i \quad\Longrightarrow\quad \Delta\Omega_i = \frac{1}{2Hs + D}\Big(\Delta P_{mi} - \frac{\Delta P_{ei}}{k_{Si}}\Big). \tag{11.5}$$

(The neglected term $-P_{mi}^0\Delta\omega_i$ of $\frac{P_m - P_e}{\omega}$ vanishes because $P_m^0 - P_e^0/k_S = 0$.)

Parameter values (model workspace)

Values from [1, Tables 2, 5–7].

Symbol	Value	Symbol	Value
$S_1,\ S_2,\ S_g$	1500, 4000, 1500 MVA	$H,\ D$	6.3 s, 2
$x_d,\ x_q$	2.135, 2.046 pu (own)	$x'_d,\ T'_{d0}$	0.34 pu, 6.47 s
$T_R,\ K_A$	0.01 s, 200	$T_B,\ T_C$	10 s, 1 s
$V_{R\max},\ V_{R\min}$	999, −999	x_g	$0.15/k_{Sg}$
$K = 1/R$	20	T_3	0.1 s
$U_o,\ U_c$	0.5, −0.5 pu/s	$P_{\max},\ P_{\min}$	1, 0 pu
$T_4,\ T_5,\ T_6$	0.3, 7, 0.6 s	$K_1,\ K_3,\ K_5$	0.3, 0.3, 0.4
R_g	0.05	$K_w,\ T_w$	3, 1 s
T_f	0.02 s	K_i	0.1
P_{SG1}	1350 MW	$P_{L2},\ P_{L3}$	1250, 4000 MW
ΔP_{L2}	−100 MW at $t_{\text{step}} = 1$ s	t_{stop}	100 s
Resulting operating point (from the load flow and initialisation above):			
$\lvert\bar E_{Q1}\rvert,\ \delta_1^0$	2.2296, 1.0923 rad	$\lvert\bar E_{Q2}\rvert,\ \delta_2^0$	2.3492, 1.0144 rad
$I_{d1}^0,\ E'^0_{q1}$	0.8142, 0.8406	$I_{d2}^0,\ E'^0_{q2}$	0.8901, 0.8307
$E_{fd1}^0,\ V_{\text{ref},1}$	2.3021, 1.01151	$E_{fd2}^0,\ V_{\text{ref},2}$	2.4284, 1.01214
$\lvert E_g\rvert,\ \delta_g^0$	0.9904, −0.0162 rad	$P_{m1}^0,\ P_{m2}^0$	0.900, 0.975 pu
P_{tie}^0	1.00 pu (100 MW)	$B_1,\ B_2$	$300 + 300\gamma_g$, 800
$\beta_1,\ \beta_2$	$330 + 300\gamma_g$, 880		

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